Practice 1 : Linear Regression on a Simple Dataset

You are given a dataset containing two columns:

- **X - the input variable**
- **Y - the target variable that follows an approximate linear relationship of the form:**

$$\boldsymbol{Y = 2X + 1}$$

Using this dataset (PARTBQ1.csv), perform the following tasks:

1. Load the dataset (PARTBQ1.csv) using pandas (pd.read_csv()).
2. Display the first 5 rows of the dataset.
3. Check for missing values and handle them appropriately.
4. Split the dataset into training and testing sets (70% training, 30% testing).
5. Train a Linear Regression model to predict Y from X.
6. Predict Y values for the test set.
7. Evaluate the model performance using metrics such as Mean Squared Error (MSE) and R^2 score.

Practice 2 : Step-by-Step Walmart Sales Prediction Using Linear Regression

Dataset Description

One of the leading retail stores in the US, Walmart, would like to predict the sales and demand accurately. There are certain events and holidays which impact sales on each day. There are sales data available for 45 stores of Walmart. The business is facing a challenge due to unforeseen demands and runs out of stock sometimes, due to the inappropriate machine learning algorithm. An ideal ML algorithm will predict demand accurately and ingest factors like economic conditions including CPI, Unemployment Index, etc.

Walmart runs several promotional markdown events throughout the year. These markdowns precede prominent holidays, the four largest of all, which are the Super Bowl, Labour Day, Thanksgiving, and Christmas. The weeks including these holidays are weighted five times higher in the evaluation than non-holiday weeks. Part of the challenge presented by this competition is modeling the effects of markdowns on these holiday weeks in the absence of complete/ideal historical data. Historical sales data for 45 Walmart stores located in different regions are available.

- Store - the store number
- Date - the week of sales
- Weekly_Sales - sales for the given store
- Holiday_Flag - whether the week is a special holiday week 1 – Holiday week 0 – Non-holiday week
- Temperature - Temperature on the day of sale
- Fuel_Price - Cost of fuel in the region

- CPI – Prevailing consumer price index
- Unemployment - Prevailing unemployment rate

Make sure the dataset (Walmart.csv) is loaded using pd.read_csv() before you begin.

1. Display basic information about the dataset using built-in methods.
2. Check the shape of the dataset (i.e., number of rows and columns).
3. Display data types of each column.
4. Drop unnecessary columns.
5. Check for any missing values in the dataset and fill them using the mean values.
6. Split into inputs (X) and output (Y)
7. Select features using p-values (with threshold 0.05)
8. Split into train/test sets (70/30 holdout split)
9. Build the regression model (Use Linear Regression method).
10. Evaluate your model appropriate metrics.

Note:

You must perform **at least three trials** to analyse the effect of changing parameters or preprocessing decisions on model performance.
In each trial, you can vary one or more of the following:

- The value of random_state used during data splitting.
- The method of handling missing values (e.g., **drop** missing rows vs. **fill with mean**).
- Whether to apply **feature selection** before training or not.

For each trial, report and compare your **evaluation metrics** (MSE, R^2, etc.) and include a **brief analysis** discussing how each change affects the model results.